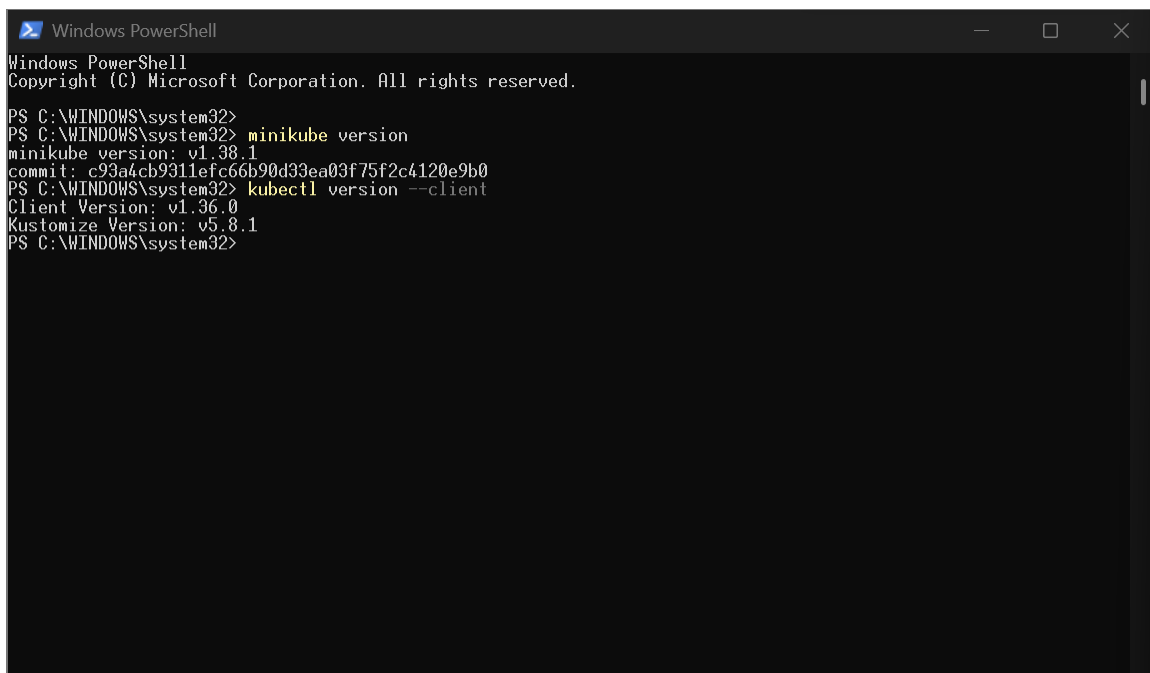


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Kubernetes

1. Install **Docker Desktop** and make sure it is running.
2. Open **PowerShell as Administrator**.
3. Install **Minikube** using:
winget install Kubernetes.minikube
4. Check Minikube using:
minikube version
5. Install **kubectl** using:
winget install -e --id Kubernetes.kubectl
6. Check kubectl using:
kubectl version --client



```
Windows PowerShell
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PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
PS C:\WINDOWS\system32> kubectl version --client
Client Version: v1.36.0
Kustomize Version: v5.8.1
PS C:\WINDOWS\system32>
```

7. Start Minikube using: minikube start --driver=docker

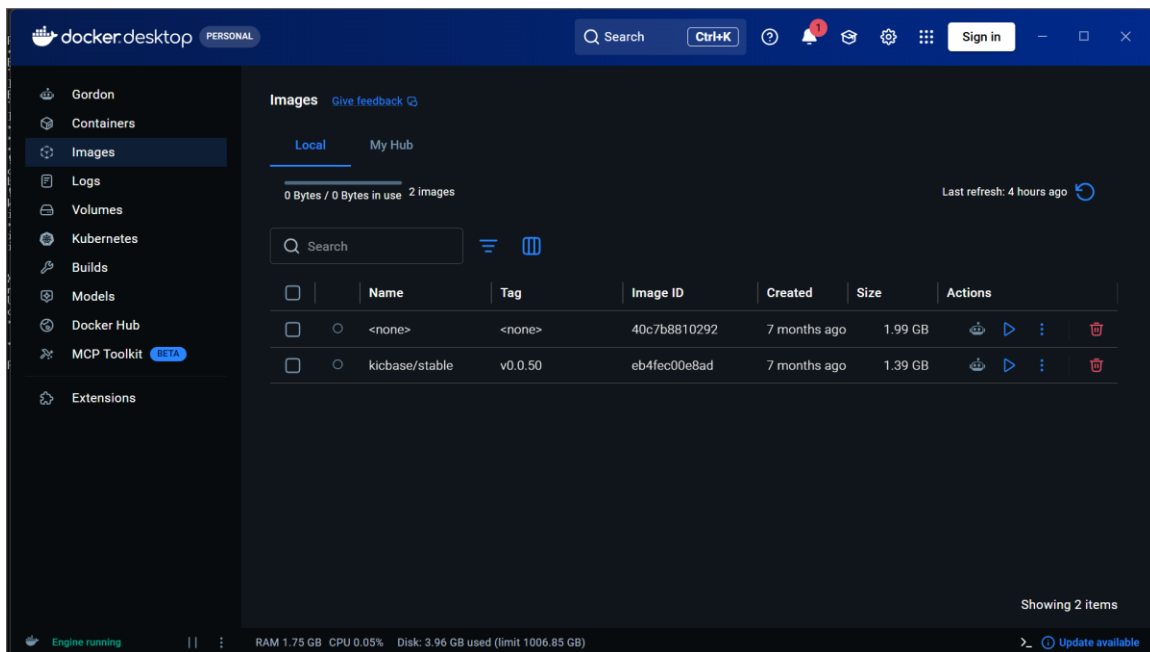
```
Select Windows PowerShell
PS C:\WINDOWS\system32> minikube start --driver=docker
* minikube v1.38.1 on Microsoft Windows 11 Home Single Language 25H2
* Using the docker driver based on user configuration
* minikube 1.39.0 is available! Download it: https://github.com/kubernetes/minikube/releases/tag/v1.39.0
* To disable this notice, run: 'minikube config set WantUpdateNotification false'

! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
  > gcr.io/k8s-minikube/kicbase...: 1.27 MiB / 519.58 MiB  0.24% 1.21 KiB p/
  > index.docker.io/kicbase/stable...: 10.08 MiB / 519.58 MiB  1.94% 13.72 KiB
  > gcr.io/k8s-minikube/kicbase...: 41.05 MiB / 519.58 MiB  7.90% 183.46 KiB
! minikube cannot pull kicbase image from any docker registry, and is trying to download kicbase tarball from gi
thub release page via HTTP.
! It's very likely that you have an internet issue. Please ensure that you can access the internet at least via
HTTP, directly or with proxy. Currently your proxy configuration is:

  > kicbase-v0.0.50-amd64.tar: 1.28 GiB / 1.28 GiB  100.00% 3.83 MiB p/s 5m4
! minikube was unable to download gcr.io/k8s-minikube/kicbase:v0.0.50, but successfully downloaded kicbase/stabl
e:v0.0.50 as a fallback image
* Creating docker container (CPUs=2, Memory=4000MB) ...
* Stopping node "minikube" ...
* Powering off "minikube" via SSH ...
* Deleting "minikube" in docker ...
! StartHost failed, but will try again: creating host: create: creating: prepare kic ssh: unable to execute icac
ls to set permissions: : exit status 1
* Failed to start docker container. Running "minikube delete" may fix it; error loading existing host. Please tr
y running [minikube delete], then run [minikube start] again; filestore "minikube": open C:\Users\ykran\.minikub
e\machines\minikube\config.json: The system cannot find the file specified.

X Exiting due to GUEST_NOT_FOUND: Failed to start host; error loading existing host. Please try running [minikub
e delete], then run [minikube start] again; filestore "minikube": open C:\Users\ykran\.minikube\machines\minikub
e\config.json: The system cannot find the file specified.
* Suggestion: minikube is missing files relating to your guest environment. This can be fixed by running 'miniku
be delete'
* Related issue: https://github.com/kubernetes/minikube/issues/9130

PS C:\WINDOWS\system32> 0_
```

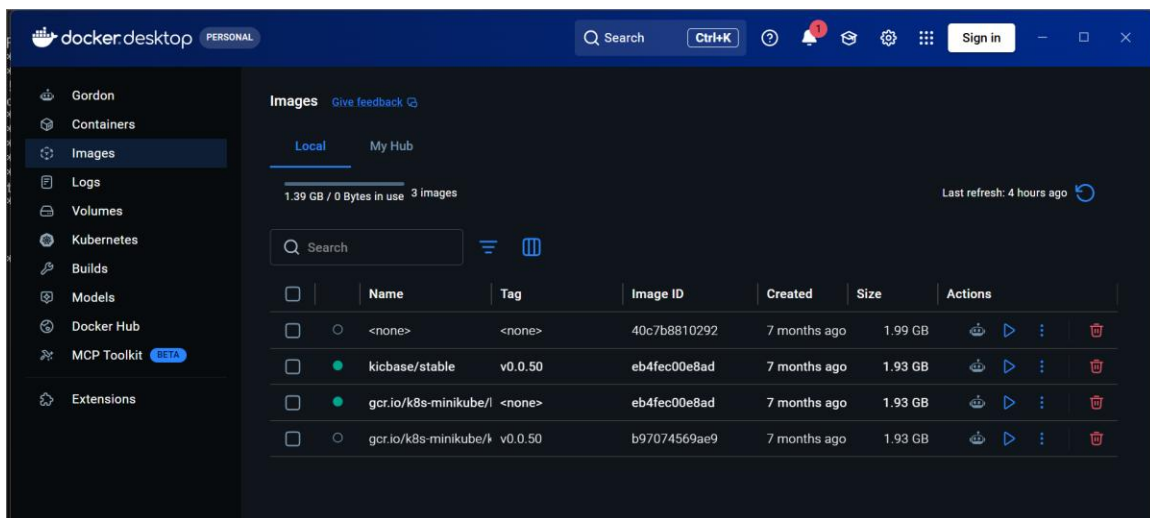


```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> minikube delete --all --purge
* Deleting "minikube" in docker ...
* Removing C:\Users\ykran\.minikube\machines\minikube ...
* Removed all traces of the "minikube" cluster.
* Successfully deleted all profiles
* Successfully purged minikube directory located at - [C:\Users\ykran\.minikube]
* Kicbase images have not been deleted. To delete images run:
  - docker rmi docker.io/kicbase/stable:v0.0.50
PS C:\WINDOWS\system32> minikube profile list

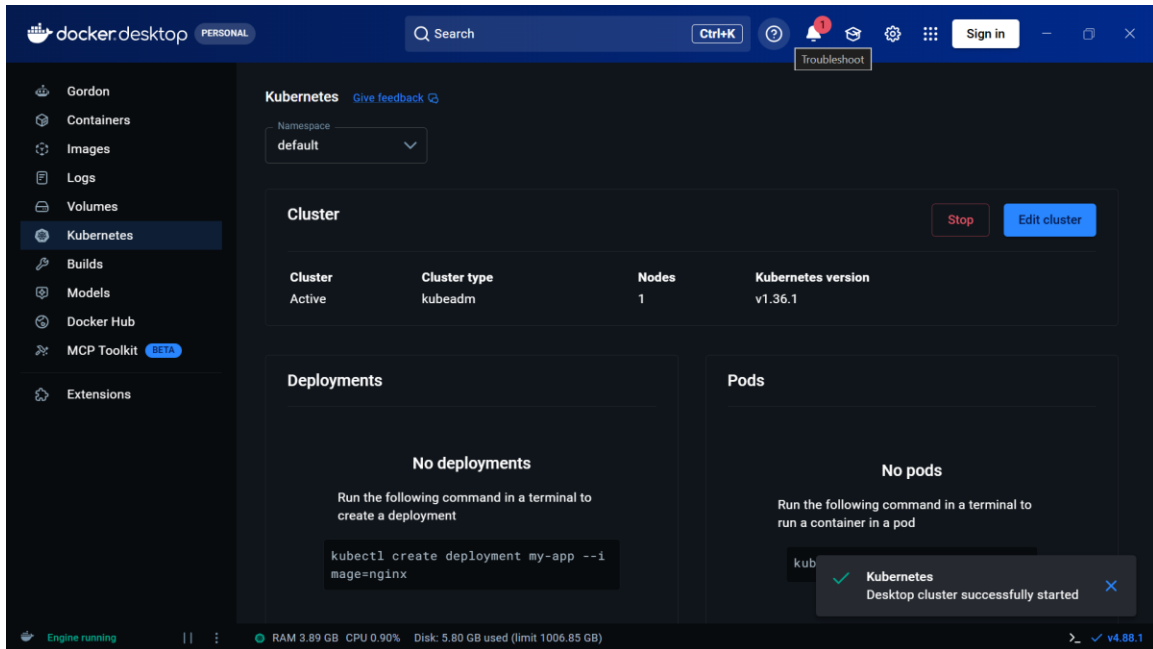
* Exiting due to MK_USAGE_NO_PROFILE: No minikube profile was found.
* Suggestion:

    You can create one using 'minikube start'.

PS C:\WINDOWS\system32>
```

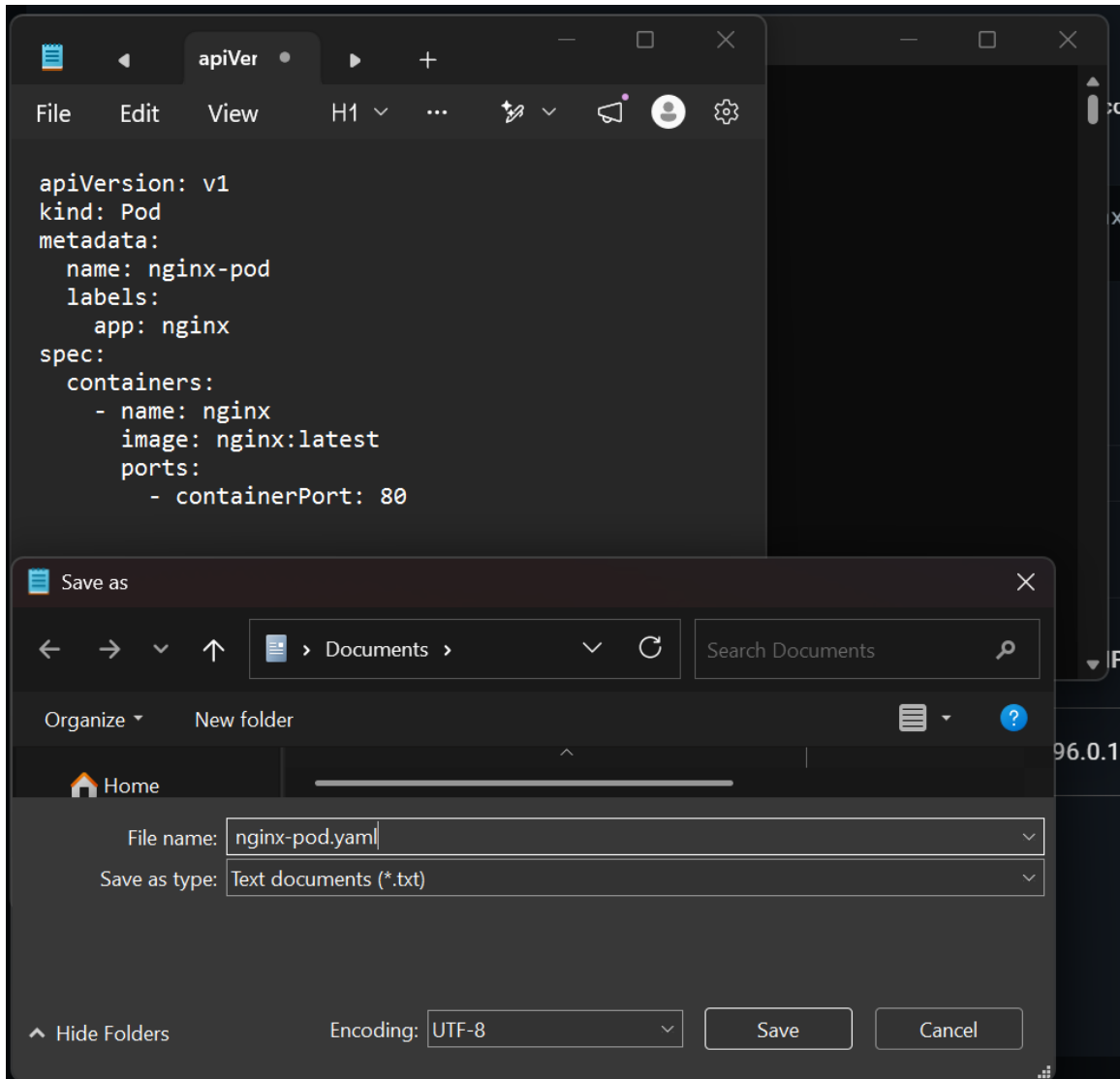


8. Check the cluster status using:
minikube status



9. Check the Kubernetes node using:
kubectl get nodes

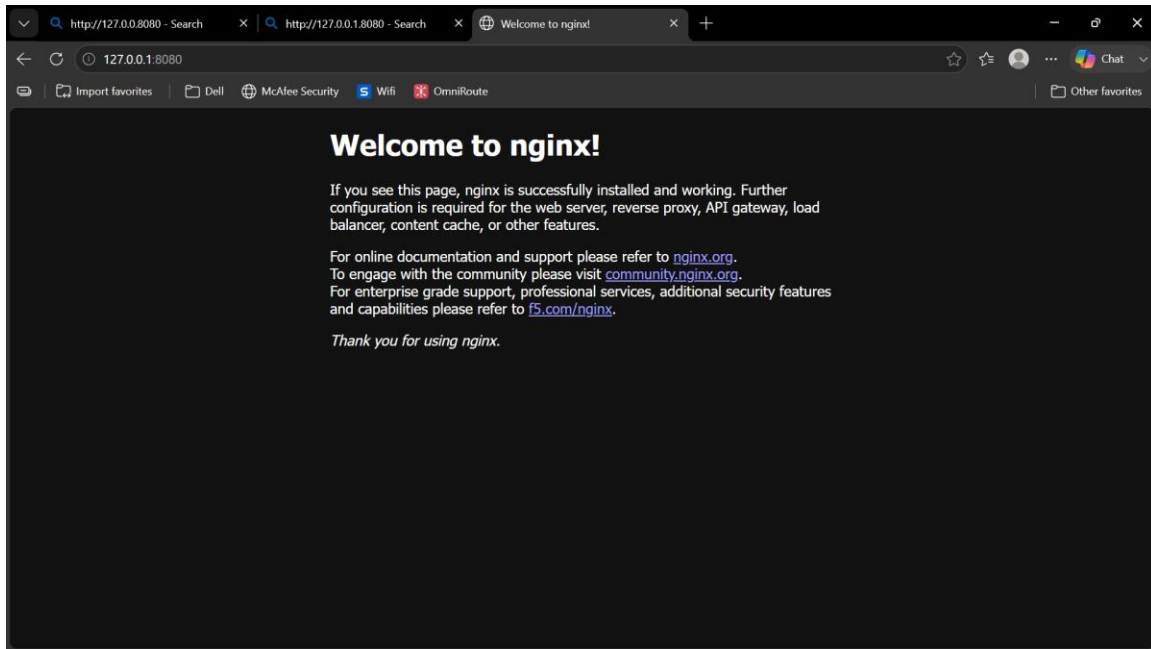
10. Create a file named **nginx-pod.yaml** and add the Nginx Pod configuration.



11. Deploy the Nginx Pod using:
`kubectl apply -f nginx-pod.yaml`

```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> kubectl config use-context docker-desktop
Switched to context "docker-desktop".
PS C:\WINDOWS\system32> kubectl get nodes
NAME                STATUS    ROLES    AGE     VERSION
docker-desktop      Ready    control-plane   4m17s   v1.36.1
PS C:\WINDOWS\system32> cd $HOME\Desktop
PS C:\Users\ykran\Desktop> kubectl apply -f nginx-pod.yaml
error: the path "nginx-pod.yaml" does not exist
PS C:\Users\ykran\Desktop> cd $HOME\Documents
PS C:\Users\ykran\Documents> kubectl apply -f nginx-pod.yaml
pod/nginx-pod created
PS C:\Users\ykran\Documents>
```

12. Check the Pod status using:
kubectl get pods
13. View Pod details using:
kubectl describe pod nginx-pod
14. Access Nginx using:
kubectl port-forward pod/nginx-pod 8080:80
15. Open <http://localhost:8080> in the browser.
16. Verify that the **Welcome to nginx!** page is displayed.



17. View Nginx logs using:
kubectll logs nginx-pod
18. Delete the Pod using:
kubectll delete -f nginx-pod.yaml
19. Stop Minikube using:
minikube stop